Professional Data Science Report

Complete Preprocessing, Leakage Controls and
Chronological Model Evaluation for Paris Apartment Valuation

Badr Ettousy

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Abstract

This report documents the complete preprocessing and empirical evaluation of a Paris apartment automated valuation model (AVM). The fixed analytical cohort contains 143,009 ordinary apartment sales from 2021–2025. Five feature layers are implemented: property characteristics, transaction time, spatial identity, strictly historical comparable sales, and external building/energy/neighborhood enrichment. The resulting Phase 2 table has 60 columns and a 44-input model contract; the enriched Phase 3 table has 141 columns and a 100-input selected model contract. Historical comparable features use only transactions at least 90 days old. DPE records are accepted only when their diagnostic date is no later than the sale date. Model development uses 2021–2023, configuration selection uses 2024, and 28,330 transactions from 2025 form the untouched final test cohort. The selected CatBoost model reaches EUR 93,197 mean absolute error (MAE), EUR 192,809 root mean squared error, $R^2 = 0.8679$, and 11.70% median absolute percentage error. It reduces MAE by EUR 3,231 (3.35%) relative to the frozen Phase 2 comparable correction; the paired 95% bootstrap interval is EUR 2,685–3,730. The gain is positive in every month and arrondissement, but calibration diagnostics show overprediction at the cheapest decile and underprediction at the most expensive decile. The model is therefore a strong automated indication, not a substitute for a certified appraisal.

143,009	**100**	**EUR 93,197**	**3.35%**
audited sales	selected inputs	2025 MAE	MAE reduction

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1 Executive assessment

The complete preprocessing and retraining workflow succeeds technically and empirically. The Gold transaction identity is unchanged across feature layers: 143,009 unique mutations enter Phase 2, and exactly the same ordered mutations leave Phase 3. All 89 repository tests and 101 artifact subtests pass after retraining, including dedicated model-evaluation and report-contract checks.

The main result is statistically consistent but deliberately described as incremental. Phase 3 reduces average absolute error from EUR 96,428 to EUR 93,197 on the same 28,330 sales. The EUR 3,231 paired reduction has a 95% bootstrap interval of EUR 2,685–3,730. Accuracy within 20% increases from 71.69% to 73.40%, while R^2 increases from 0.8601 to 0.8679. These gains support the added data engineering; they do not imply causal effects from DPE, transport or building characteristics.



Figure 1: The implemented five-part preprocessing contract and its principal quality, leakage and evaluation safeguards.

Decision summary

- **Proceed with the Phase 3 model as the research benchmark.** It dominates the frozen Phase 2 reference on MAE, MdAPE, within-20% accuracy and R^2.
- **Retain the Phase 2 correction as an operational fallback.** It is simpler, transparent and uses only DVF plus lagged market evidence.
- **Do not describe all enrichment as point-in-time.** Dated DPE and historical comparables are temporally safe; current BAN/BDNB, transport, school, park, service and noise snapshots are retrospective-static context.
- **Monitor price-extreme calibration.** The lowest-value decile is overpredicted by EUR 34,507 on average; the highest-value decile is underpredicted by EUR 221,745.
- **Seek prospective confirmation.** Aggregate 2025 performance was visible during earlier project phases even though Phase 3 fitting and selection do not use 2025 rows.

2 Problem definition and analytical cohort

The target is total transaction value, `valeur_fonciere`, for an ordinary apartment sale in Paris. The unit of analysis is a mutation, not an individual DVF property line. This distinction prevents a multi-line sale from duplicating its total price and receiving excessive training weight.

2.1 Eligibility and quality filters

The preprocessing retains transactions satisfying all of the following:

- mutation nature is *Vente*;
- exactly one apartment is present and no house or commercial unit is included in the same mutation;
- transaction date, value, built surface, Paris postcode and coordinates are present;
- built surface is 9–300 m^2;
- total value is EUR 50,000–10,000,000;
- observed price per m^2 is EUR 2,000–30,000;
- postcode is 75001–75020 and coordinates lie inside fixed Paris bounds.

The current transaction's price per m^2 is calculated only for filtering and as the source value from which *future transactions* may obtain historical comparable statistics. It is explicitly excluded from the current row's model inputs.

2.2 Chronological split

Table 1: Fixed chronological cohort and evaluation role.

Year	Transactions	Cumulative train size	Role
2021	30,397	30,397	Development
2022	33,080	63,477	Development
2023	26,798	90,275	Development
2024	24,404	114,679	Configuration selection
2025	28,330	—	Frozen final test
Total	143,009	—	Fixed Gold identity

No random train/test split is used. Hyperparameters and feature families are selected on 2024 only. The chosen configuration is then refitted on 2021–2024 and evaluated once on 2025. This rolling-origin design is materially closer to deployment than a random split in a changing property market.

3 Five-part feature engineering

3.1 1. Property representation

The core property layer contains built surface, principal-room count, address number, lot count and available terrain surface. Two derived quantities improve scale representation: $\log(1+\text{surface})$ and rooms per 10 m^2. Surface remains both raw and transformed because tree models can exploit different threshold structures. Missing optional room, lot and terrain values remain explicitly missing for model-native handling; mandatory surface is never missing after filtering.

3.2 2. Transaction time

Calendar variables are month, day of year, year and months elapsed since January 2021. These variables expose seasonality and broad price-level drift without using future target aggregates. They are known at valuation time and therefore satisfy the point-in-time contract.

3.3 3. Location and spatial identity

WGS84 longitude/latitude are transformed to Lambert-93 coordinates (EPSG:2154), allowing Euclidean distances in metres. The pipeline adds distance from central Paris and sine/cosine bearing. H3 cells at resolutions 8, 9 and 10 support multiscale spatial identity. Resolutions 8 and 9 enter the benchmark; resolution 10, street, address and parcel IDs remain in Gold for audit and history rather than unrestricted memorization.

3.4 4. Historical comparable sales

For transaction i at date t_i, an eligible comparable j must satisfy

$$t_i - t_j \geq 90 \text{ days}, \qquad t_i - t_j \leq w, \qquad d(i,j) \leq r,$$

where (r, w) spans seven windows: 250 m/365 days; 500 m/180, 365 and 730 days; and 1,000 m/180, 365 and 730 days. Each window produces a count and median price per m^2. Additional variables measure median absolute deviation, nearest prior sale distance, median comparable age, effective sample size and 180-versus-365 day price momentum.

Distance and time-decayed comparable price is

$$\widehat{p}_i = \frac{\sum_{j \in \mathcal{C}_i} p_j \exp(-d_{ij}/350) \exp(-(a_{ij} - 90)/240)}{\sum_{j \in \mathcal{C}_i} \exp(-d_{ij}/350) \exp(-(a_{ij} - 90)/240)}.$$

A second estimator restricts candidates to 70–130% of the target surface and within one principal room when room information exists. The feature builder was explicitly tested to ensure the current transaction is never its own comparable and that empty historical neighborhoods produce valid zero counts, not failed quality reports.

Weighted-comparable coverage is 77.64% in the initial 2021 cohort because no earlier project-year history exists. Coverage reaches 100% from 2022 onward; the median 2025 transaction has 243 prior sales within 500 m/365 days and 836 within 1,000 m/365 days.

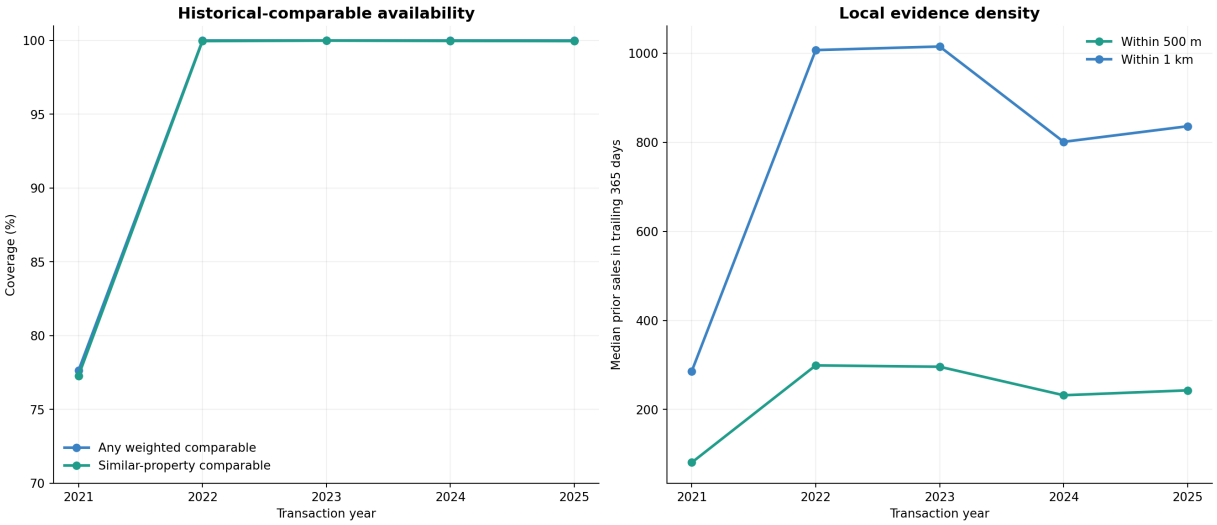


Figure 2: Historical comparable coverage and density across transaction years.

3.5 5. Building, energy and neighborhood enrichment

The external layer links DVF transactions to BAN address identity and BDNB building identity, then adds physical building attributes, point-in-time DPE, transport accessibility, schools, parks, services and strategic noise maps. Exact BAN address resolution covers 99.49% of transactions; a BDNB building is resolved for 99.997%, and 99.46% receive a high-confidence BDNB match.

Building features include construction year/period, age at sale, level count, height, footprint, dwelling and lot counts, wall/roof materials, usage and coarse state. DPE includes energy/GHG classes, consumption, emissions, heating energy, habitable surfaces, levels and diagnostic-match confidence. Context features contain distance and density measures for metro, RER, train, tram, schools, green spaces and services, plus road/RATP/SNCF noise bands.

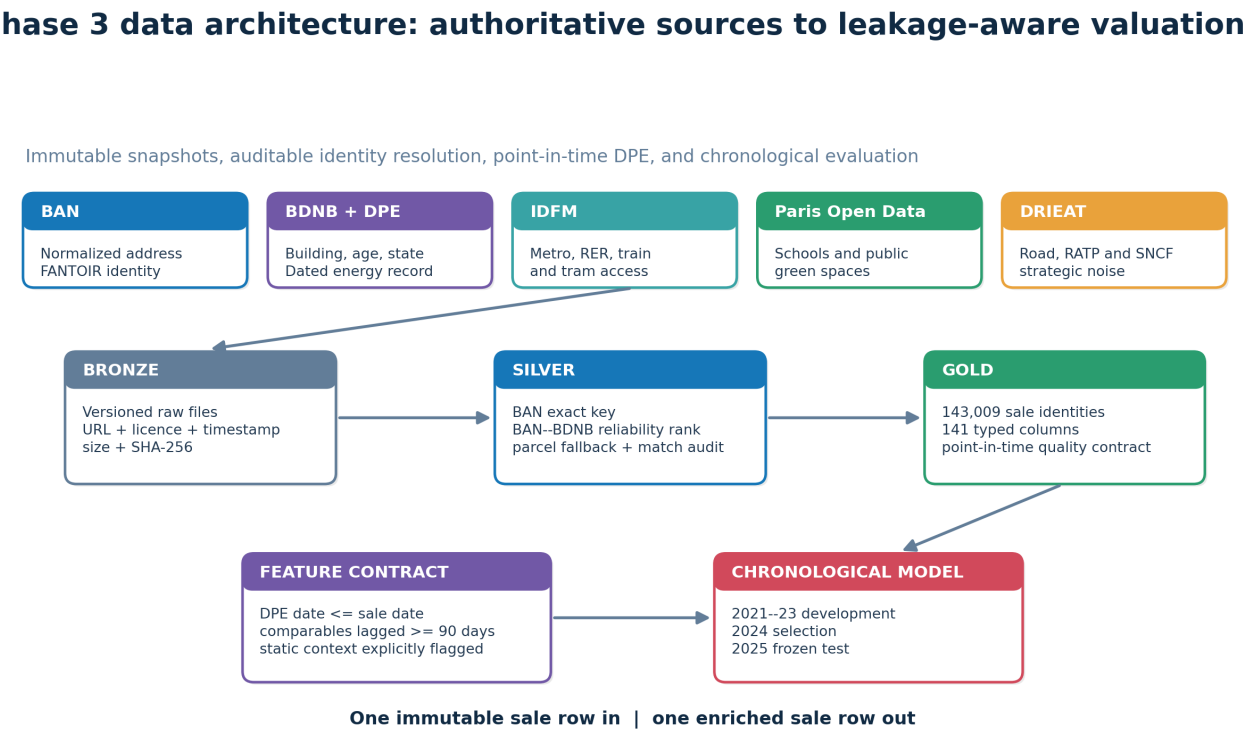


Figure 3: Bronze–Silver–Gold enrichment architecture. Entity-resolution audit fields make one-to-many source relationships explicit.

4 Temporal integrity, provenance and validation

4.1 Point-in-time versus retrospective-static fields

Two feature families satisfy strict historical availability:

1. historical comparable features, with a minimum 90-day availability delay; and
2. DPE observations whose `date_etablissement_dpe` is no later than `date_mutation`.

The completed table contains zero future-DPE joins.

BAN/BDNB identity and physical attributes, current transport/school/park/service locations and the 2022 strategic noise map are retrospective-static context. They are appropriate for the stated reconstruction experiment but must not be presented as if every value were available at each historical sale date.

4.2 Artifact controls

Each stage publishes atomically: a temporary file is fully written before it replaces the previous valid artifact. Quality reports record source and output SHA-256 checksums, row counts, date range, coordinate reference systems, feature names, comparable coverage, join coverage and missingness. Phase 3 asserts that enrichment preserves row count, order and mutation identity.

The unified command is:

```
python -m paris_avm.features.preprocessing
```

If enrichment is interrupted after Phase 2, `-reuse-phase2` verifies the existing Phase 2 checksum before resuming, avoiding an unnecessary comparable-feature rebuild.

5 Modeling and statistical protocol

5.1 Phase 2 candidates

Phase 2 evaluates LightGBM and XGBoost on total value, price per m^2 and a residual target, as well as spatial-only and comparable-only ablations. A validation-selected ensemble is also evaluated. The winning Phase 2 method is the transparent correction

$$\widehat{V}_i = 0.75\widehat{V}_i^{\text{LightGBM}} + 0.25\left(\widetilde{p}_{i,365d,500m} \times S_i\right),$$

where S_i is apartment surface and the local median falls back to the base prediction when unavailable. It obtains EUR 96,428 MAE and becomes the frozen reference for Phase 3.

5.2 Phase 3 candidates and selection

LightGBM ablations isolate Phase 2, dated DPE, building and neighborhood families, then combine all enrichment without raw external identity. Both MAE and Huber objectives are evaluated. CatBoost additionally evaluates the full 100-input contract, including three externally resolved BAN/BDNB identity fields. Selection uses 2024 validation MAE only.

The selected CatBoost configuration has depth 7, L_2 leaf regularization 5 and MAE loss. It is refitted on all 114,679 transactions from 2021–2024 before producing the 2025 predictions.

5.3 Metrics and uncertainty

Reported metrics are MAE, RMSE, R^2, mean and median absolute percentage error (MAPE/MdAPE), and fractions within 10% and 20%. MAE confidence intervals use 500 nonparametric bootstrap resamples of 2025 transactions. Improvement intervals are paired: each resample preserves the Phase 2 and Phase 3 errors of the same transaction.

6 Primary model results

Table 2: Frozen 2025 performance on 28,330 identical transactions.

Model	MAE	RMSE	R^2	MdAPE	Within 10%	Within 20%
Phase 2 comparable correction	96,428	—	0.8601	12.22%	—	71.69%
LightGBM, Phase 2 inputs	95,823	—	0.8630	—	—	—
LightGBM, neighborhood	95,760	—	0.8637	—	—	—
LightGBM, dated DPE	95,183	—	0.8634	—	—	—
LightGBM, building	94,528	—	0.8660	—	—	—
LightGBM, full no identity	94,204	—	0.8654	—	—	—
CatBoost, full 100 inputs	**93,197**	**192,809**	**0.8679**	**11.70%**	**43.86%**	**73.40%**

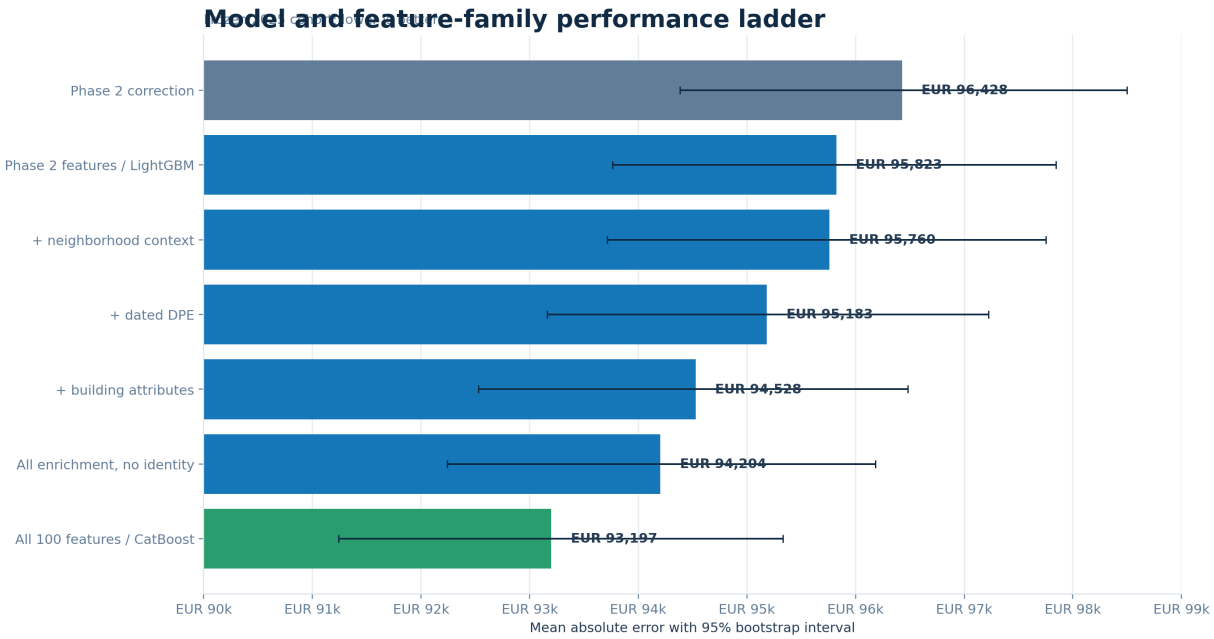


Figure 4: Selected feature-family and model ladder with 95% bootstrap intervals.

The DPE, building and context ablations each outperform the frozen Phase 2 correction, while the combined no-identity model performs better than any single family. CatBoost with all features improves further. The differences are predictive associations, not estimates of the market value caused by a specific building or neighborhood attribute.

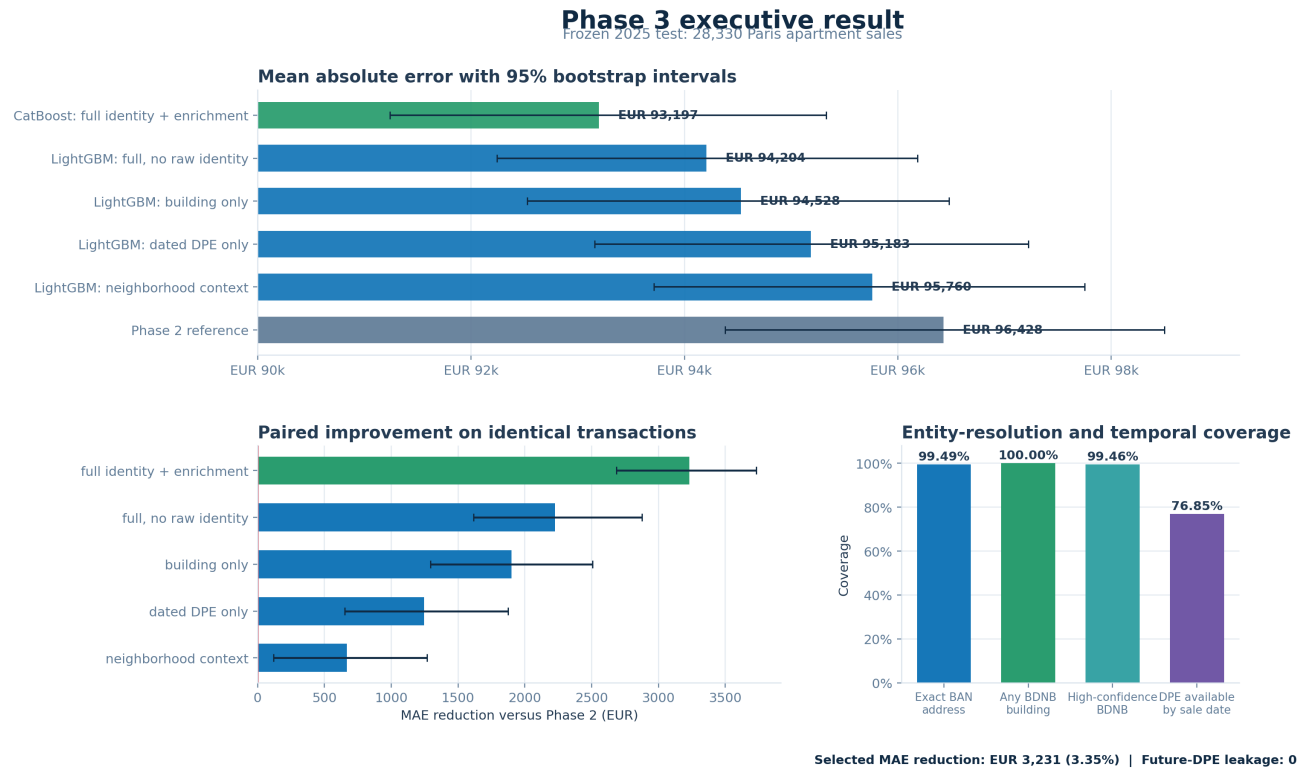


Figure 5: Executive result, paired improvement intervals and entity-resolution coverage. The selected MAE reduction is EUR 3,231 (3.35%).

7 Error analysis and operational robustness

7.1 Time and geography

Phase 3 improves MAE in every month. Monthly reductions range from EUR 1,367 (1.53%) in May to EUR 5,078 (5.11%) in October. December is the hardest month for both systems, with Phase 3 MAE of EUR 101,399; April is the lowest-error month at EUR 81,428. There is no month in which the enrichment model loses to the Phase 2 reference.

The gain is also positive in all 20 arrondissements. Absolute reductions are largest in the 6th (EUR 14,360), 4th (EUR 13,965), 8th (EUR 13,828) and 7th (EUR 11,718), where transaction values and errors are relatively high. The smallest reduction is EUR 268 in the 16th. Geography therefore changes the magnitude of benefit, not its sign on this cohort.

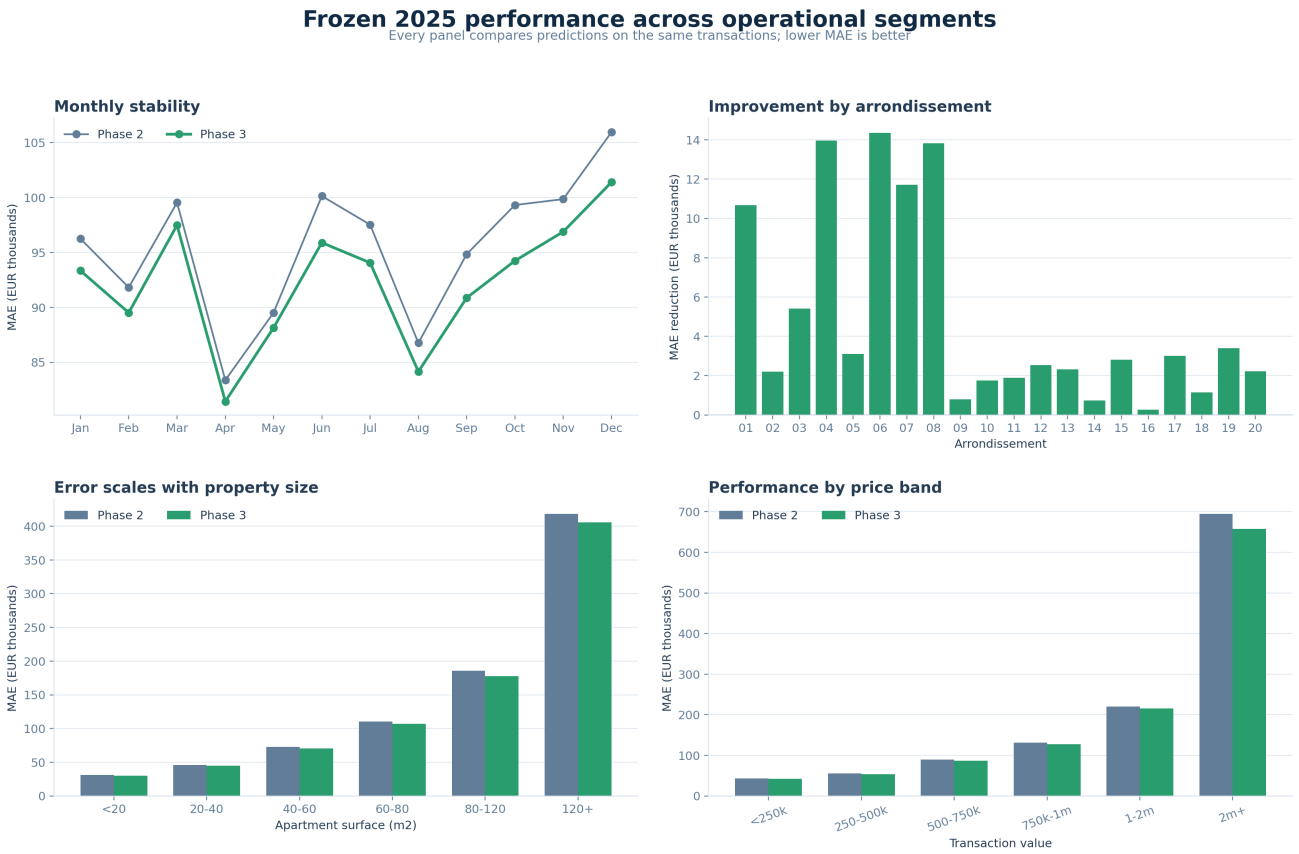


Figure 6: Paired Phase 2/Phase 3 diagnostics by month, arrondissement, apartment surface and transaction-value band.

7.2 Surface and price bands

Absolute error scales sharply with property size. Phase 3 MAE is EUR 30,122 for apartments below 20 m^2, EUR 70,564 at 40–60 m^2, EUR 177,480 at 80–120 m^2, and EUR 405,677 above 120 m^2. Enrichment improves every size band; the largest absolute reduction is EUR 12,670 for the 120+ m^2 group.

The same pattern appears by value. Phase 3 MAE is EUR 42,537 below EUR 250,000 and EUR 657,964 above EUR 2 million. The highest-value group improves most in absolute and relative terms (EUR 36,868; 5.31%) but retains a 20.43% MdAPE. High-value properties remain a distinct operational risk segment.

7.3 Comparable density and DPE availability

All 2025 transactions have historical comparable coverage under the broad 1,000 m/730-day definition. Under the tighter 500 m/365-day count, 162 sales have 1–49 comparables, 9,277 have 50–199, 18,231 have 200–499 and 660 have 500 or more. Phase 3 improves every group. The largest absolute improvement, EUR 5,026, occurs for the 50–199 group.

Point-in-time DPE is available for 28,140 of 28,330 test sales (99.33%). The remaining 190 transactions have much higher Phase 3 MAE (EUR 149,841), partly because this very small group differs in composition. Phase 3 still improves both DPE-available and unavailable groups. This comparison is descriptive and does not isolate a causal DPE contribution.

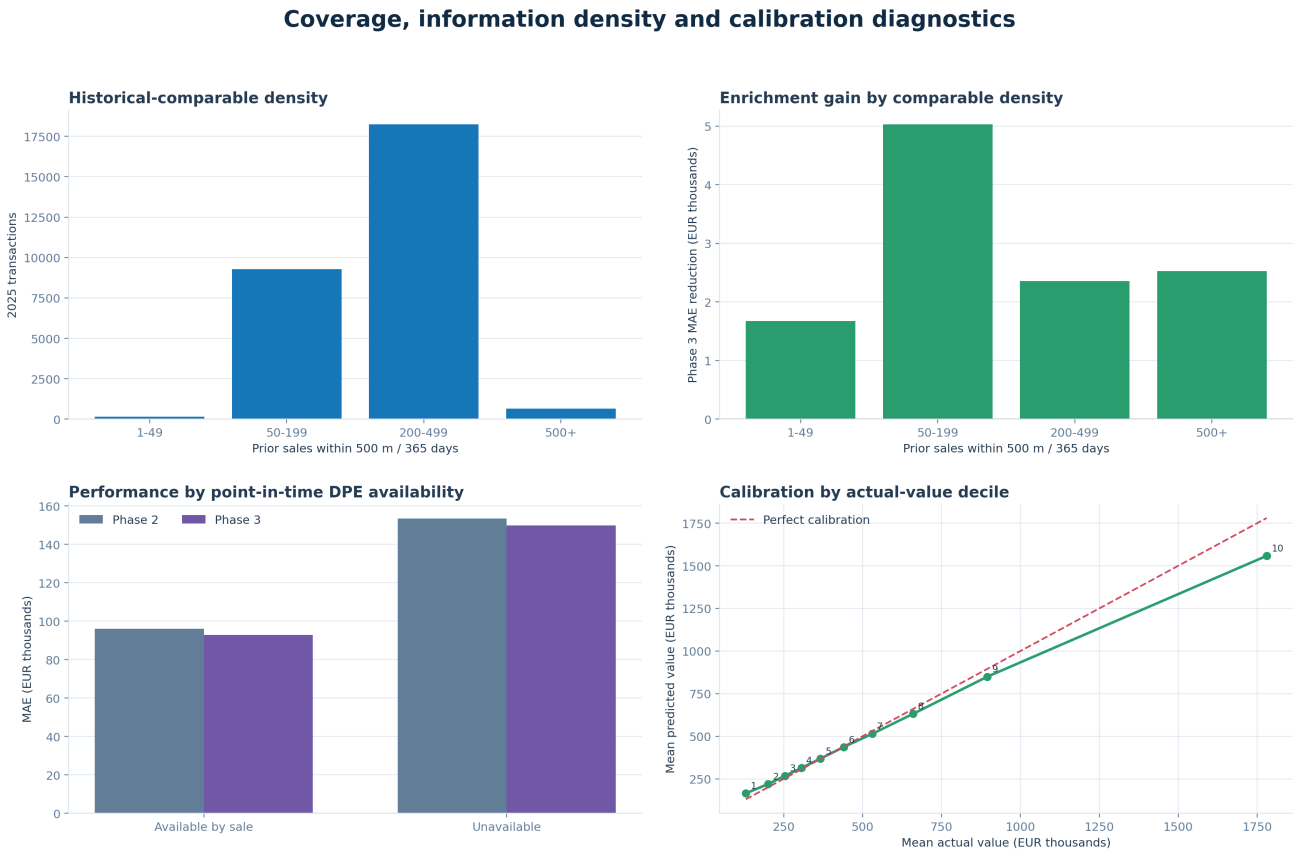


Figure 7: Comparable density, enrichment gain, DPE availability and actual-value-decile calibration on the frozen 2025 cohort.

8 Calibration, residuals and spatial behavior

Average calibration is strongest through the middle of the value distribution. The lowest decile has mean actual value EUR 130,165 and mean prediction EUR 164,672, an average overprediction of EUR 34,507. Bias changes sign near the sixth decile. The highest decile has mean actual value EUR 1,780,241 and mean prediction EUR 1,558,496, an average underprediction of EUR 221,745. This is the familiar regression-to-the-mean pattern and motivates segment-aware monitoring or a validation-fitted calibration layer.

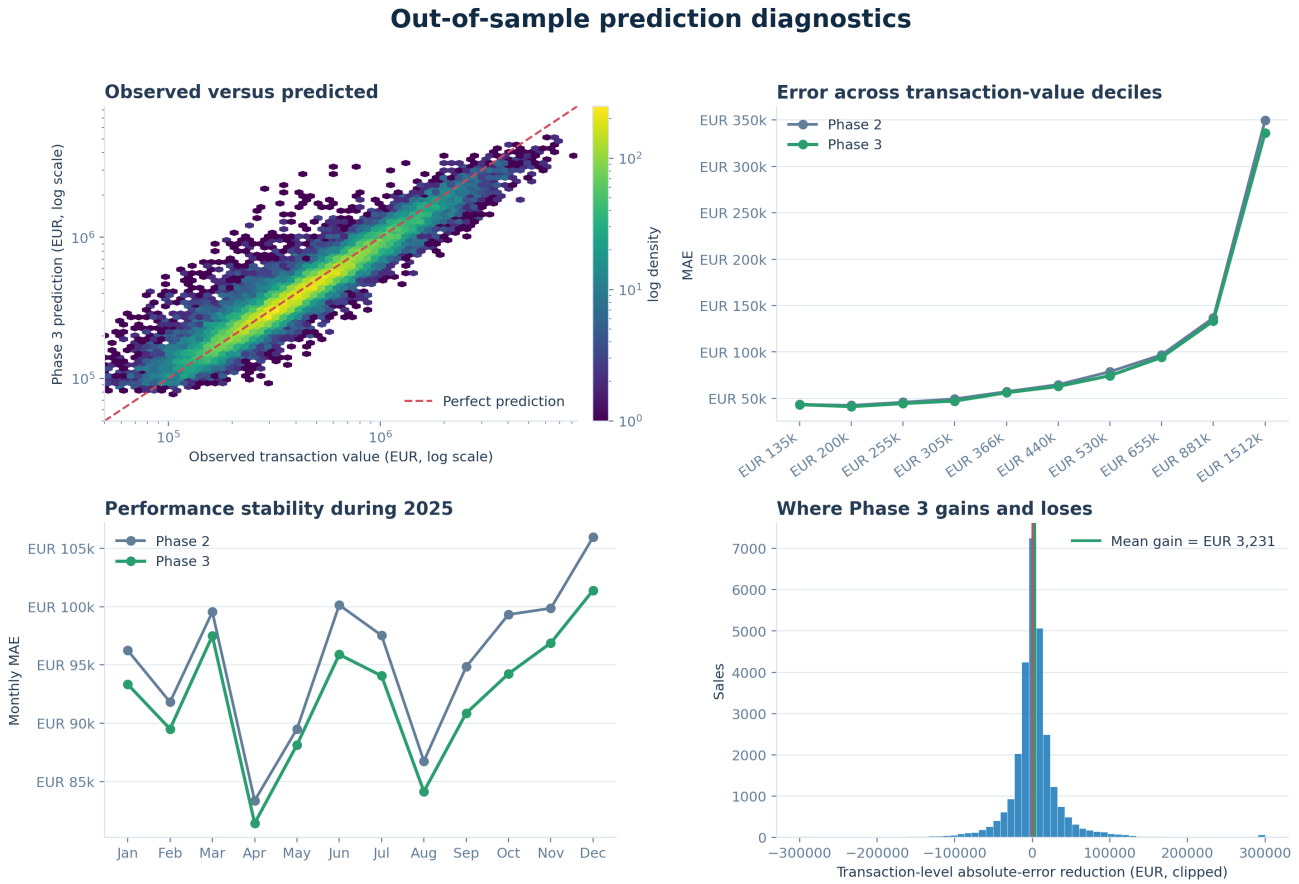


Figure 8: Out-of-sample prediction, residual, monthly and cumulative-error diagnostics for the selected model and frozen reference.

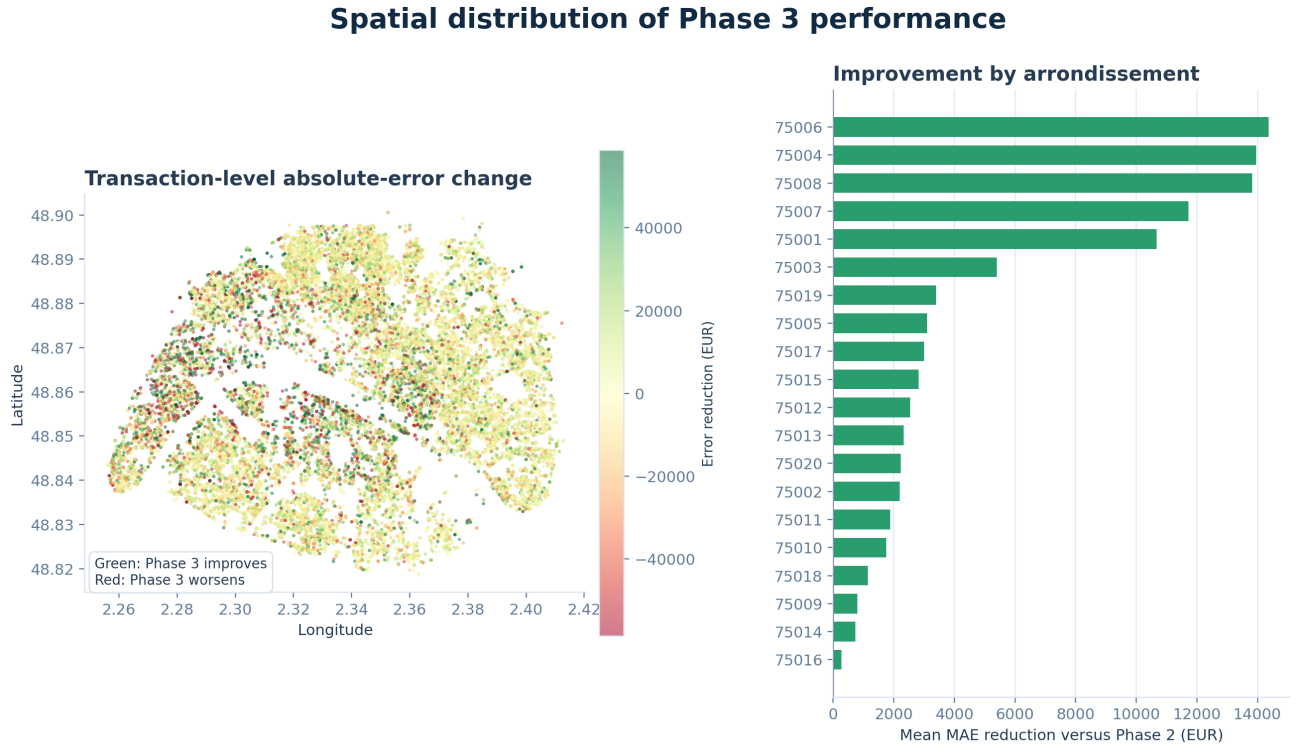


Figure 9: Spatial distribution of error and paired improvement across Paris.

9 Feature importance and interpretation

CatBoost prediction-value-change importance attributes 79.57% of total importance to core DVF/spatial inputs and 12.95% to historical comparables. Building/linkage, neighborhood context, DPE/energy and micro-location identity account for 2.39%, 2.07%, 1.86% and 1.16%, respectively. The model therefore remains primarily a surface-and-market-evidence system; enrichment supplies smaller refinements that collectively improve generalization.

The two dominant individual features are log surface (37.92%) and raw surface (35.28%). The strongest historical signals are 365-day/500-m local median price per m^2, the 365-day/250-m median, similar-comparable price and the weighted comparable estimate. DPE habitable surface is the highest-ranked DPE field. Importance is not causal and correlated features divide credit in a model-dependent manner.

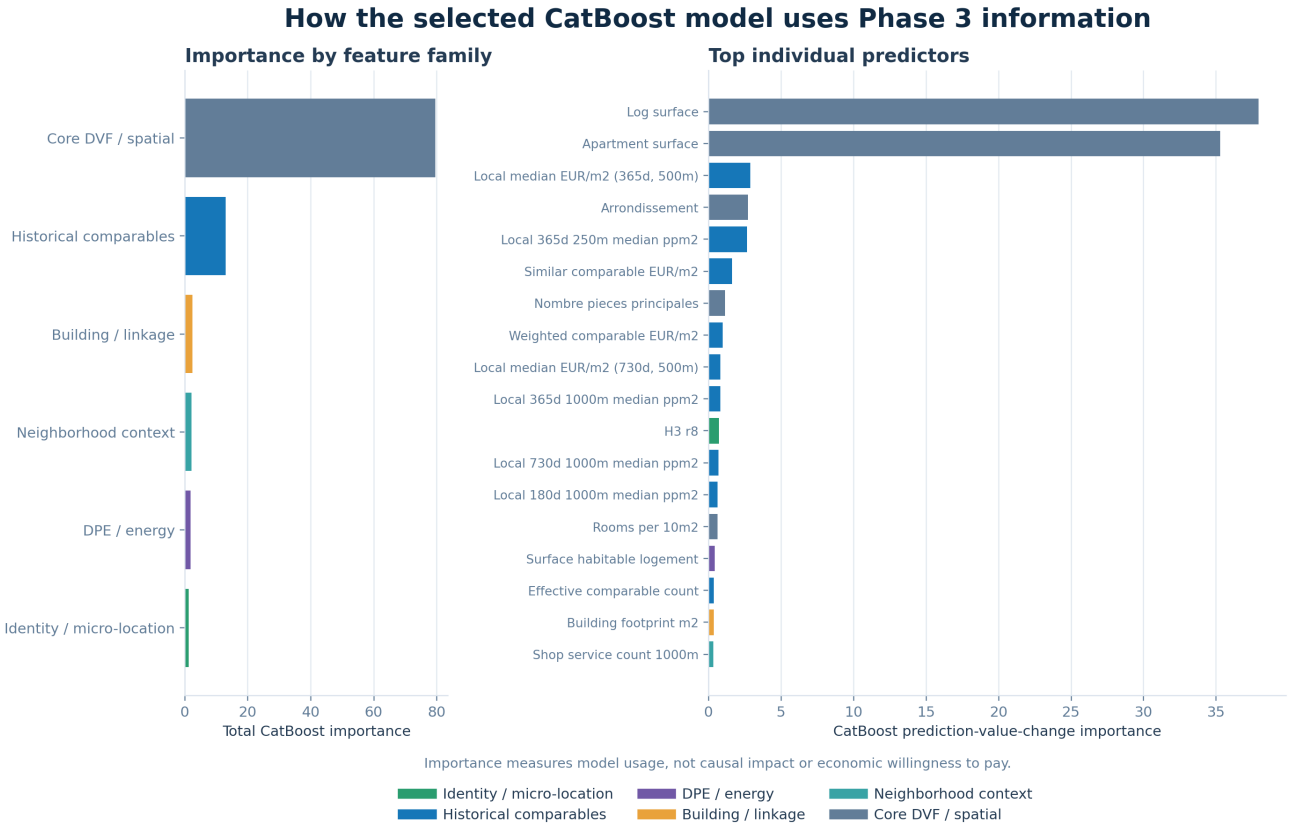


Figure 10: Selected CatBoost importance by feature family and leading individual variables.

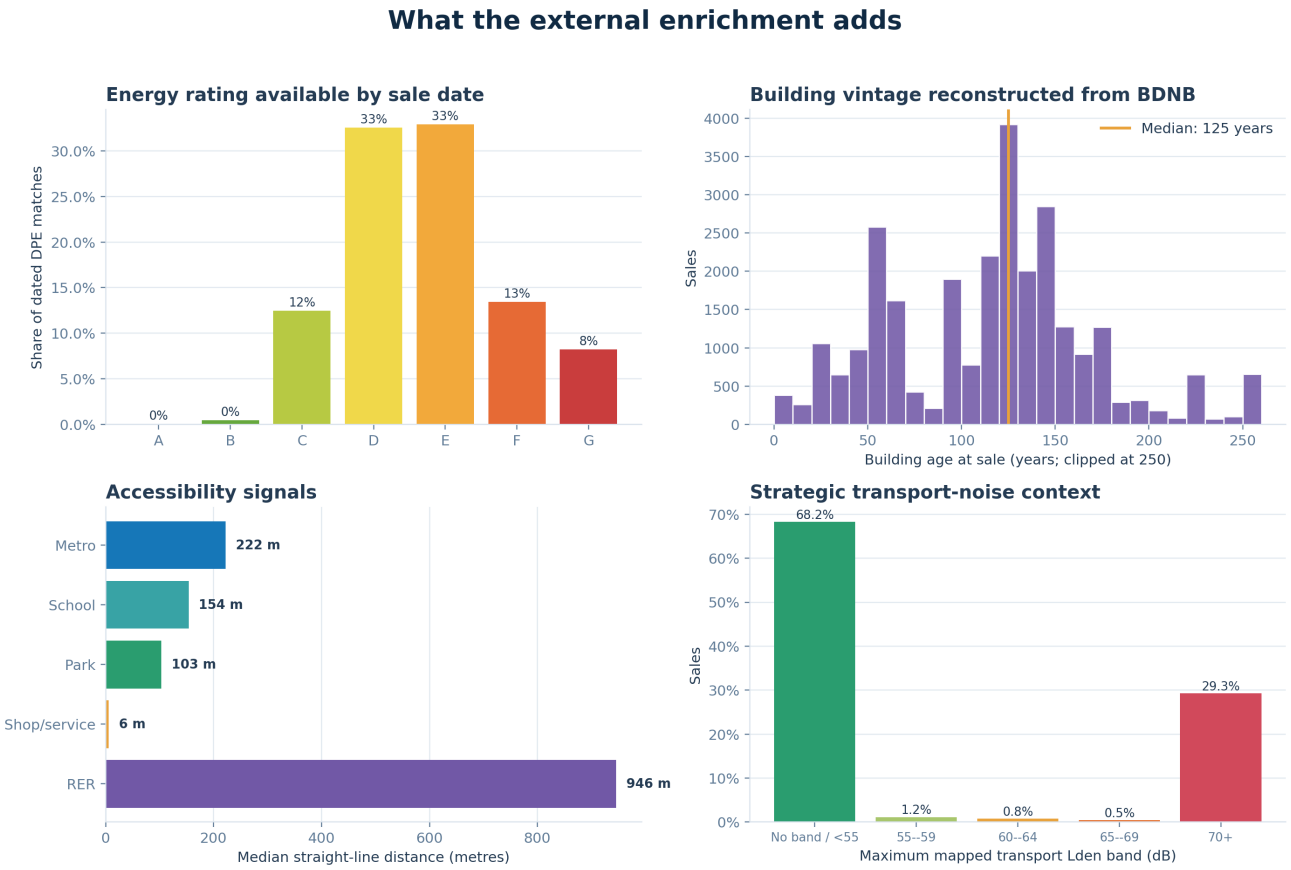


Figure 11: Coverage and distributions of the building, energy, transport, amenity and noise enrichment layer.

10 Limitations and model-risk controls

1. **Retrospective-static bias.** Current BAN/BDNB and amenity snapshots can improve reconstruction without proving that every field was known historically.

2. **Test-year familiarity.** The 2025 cohort is excluded from fitting and selection, but earlier project phases exposed aggregate 2025 scores. A later untouched cohort is needed for prospective confirmation.

3. **Missing apartment-level attributes.** Floor, elevator, interior condition, view, orientation, renovation and occupancy are unavailable.

4. **DPE linkage ambiguity.** A building can contain multiple apartments and eligible diagnostics. Confidence and surface-difference fields expose but cannot remove this ambiguity.

5. **Spatial semantics.** Straight-line distance ignores walking networks, barriers, station entrances and service quality.

6. **Market-regime uncertainty.** Transaction-level bootstrap intervals do not capture interest-rate, regulation, tax or liquidity regime shifts.

7. **Tail calibration.** The model compresses price extremes and should be monitored separately for luxury and very small/low-value units.

8. **Use limitation.** Output is an automated indication with statistical uncertainty, not a certified appraisal or lending decision.

11 Recommended production roadmap

1. Freeze this benchmark and preserve its source/output checksums.
2. Validate prospectively on the next complete DVF period without changing configuration after observing its labels.
3. Add validation-fitted calibration for price tails and publish interval coverage by arrondissement, surface and value band.
4. Acquire legally reusable apartment-floor, elevator and condition data; introduce each family through a dated ablation.
5. Monitor monthly MAE, MdAPE, bias, within-20% rate, missingness, match confidence and distribution drift.
6. Provide a fallback to the transparent Phase 2 correction when external resolution or data freshness fails.
7. Record model cards, data lineage, intended use, exclusions and human review requirements before deployment.

12 Reproducibility package

The complete workflow is reproducible from the project root:

```
$env:PYTHONPATH = '.\src'
python -m paris_avm.features.preprocessing
python -m paris_avm.modeling.benchmark_phase2
python -m paris_avm.modeling.benchmark_phase3
python -m paris_avm.visualization.phase2
python -m paris_avm.visualization.phase3
python -m paris_avm.visualization.model_evaluation
powershell -ExecutionPolicy Bypass -File `
  .\docs\paper\compile_model_evaluation.ps1
```

Primary machine-readable deliverables are:

- `reports/preprocessing/run_summary.json`;
- `reports/phase2/phase2_model_comparison.csv`;
- `reports/phase3/phase3_model_comparison.csv`;
- `reports/phase3/phase3_test_predictions.csv`;
- `reports/model_evaluation/segment_performance.csv`;
- `reports/model_evaluation/calibration_by_value_decile.csv`;
- `reports/model_evaluation/feature_importance.csv`; and
- `reports/model_evaluation/feature_family_importance.csv`.

13 Conclusion

The five proposed preprocessing layers are fully implemented, tested and run on the complete Paris cohort. The data contract is explicit, row identity is preserved, target leakage is controlled for current-sale price and dated DPE, and chronological evaluation is enforced. Retraining confirms that external building, energy and neighborhood enrichment produces a real but bounded improvement over a strong comparable-sale reference. The selected 100-input CatBoost model reduces 2025 MAE by EUR 3,231, improves every month and arrondissement, and preserves strong overall R^2. Its remaining weaknesses are concentrated in price extremes and missing apartment-level condition.

The professional conclusion is therefore balanced: the enriched model is the best research benchmark and a credible AVM component, while prospective validation, temporal source vintages, tail calibration and human appraisal controls remain necessary before high-stakes use.

A Feature-contract inventory

Table 3: Model-input inventory by family.

Family	Inputs
Property	`surface_reelle_bati`, `nombre_pieces_principales`, `adresse_numero`, `nombre_lots`, `log_surface`, `rooms_per_10m2`, `surface_terrain`
Time	`month`, `day_of_year`, `year`, `months_since_2021`
Coordinates and spatial	`latitude`, `longitude`, `x_l93`, `y_l93`, `distance_paris_center_m`, `bearing_sin`, `bearing_cos`, `h3_r8`, `h3_r9`, `code_postal`
Historical comparables	Seven local count/median pairs; 365-day/500-m MAD; weighted and similar-weighted price per m^2; similar/effective counts; nearest prior distance; median comparable age; 180-versus-365-day momentum at 500 m and 1,000 m
Building/linkage	BAN distance/confidence; BDNB confidence and candidate counts; levels, year, dwelling count, height, footprint, parking/total lots, age, year-missing flag, use, wall/roof materials, state and RNC period
DPE/energy	availability, age, surface difference, confidence, construction year, dwelling/building levels and surfaces, energy consumption, GHG emissions, DPE/GHG classes, heating installation and heating energy
Neighborhood context	metro/RER/train/tram distance; station-mode and line counts; school distance/count; green-space distance/count/area; service distance/counts; road, RATP, SNCF and maximum transport noise
External identity	`ban_address_id`, `bdnb_building_id`, `bdnb_match_method`

B Machine-readable diagnostic tables

The PDF intentionally reports interpretable aggregates rather than every CSV row. Complete results for 12 months, 20 arrondissements, six surface bands, six value bands, comparable-density bands, DPE availability and ten calibration deciles are stored in `reports/model_evaluation/segment_performance.csv` and `reports/model_evaluation/calibration_by_value_decile.csv`. These tables are generated directly from the frozen prediction export and are the source of Figures 6 and 7.

References

[1] Direction générale des Finances publiques. *Demandes de valeurs foncières (DVF)*. https://www.data.gouv.fr/fr/datasets/demandes-de-valeurs-foncieres/.

[2] Institut national de l'information géographique et forestière. *Base Adresse Nationale*. https://adresse.data.gouv.fr/.

[3] Centre scientifique et technique du bâtiment. *Base de Données Nationale des Bâtiments*. https://www.data.gouv.fr/fr/datasets/base-de-donnees-nationale-des-batiments/.

[4] ADEME. *Diagnostics de performance énergétique pour les logements*. https://data.ademe.fr/.

[5] L. Prokhorenkova et al. CatBoost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, 31, 2018.

[6] G. Ke et al. LightGBM: a highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30, 2017.